7 A sphere X collides head on with a second identical sphere Y which is stationary.

The mass of each sphere is 0.15 kg.

Sphere X is travelling at a velocity of 2.0 m / s before the collision and produces an impulse of 0.21 N s on sphere Y.

What is the velocity of sphere X after collision?

A 0.60 m / s in the opposite direction to Y

B 0.60 m / s in the same direction as Y

C 1.4 m / s in the opposite direction to Y

D 1.4 m / s in the same direction as Y